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Evolution and function of *ubiquitin-specific proteases* (*UBPs*): Insight into seed development roles in plants

Yunpeng Cao [a,c,f,*], Yanli Li [f], Lihu Wang [e], Lin Zhang [d,**], Lan Jiang [b,**]

[a] CAS Key Laboratory of Plant Germplasm Enhancement and Specialty Agriculture, Wuhan Botanical Garden, Chinese Academy of Sciences, Wuhan 430074, China
[b] Key Laboratory of Non-coding RNA Transformation Research of Anhui Higher Education Institution, Yijishan Hospital of Wannan Medical College, Wuhu 241000, China
[c] Center of Economic Botany, Core Botanical Gardens, Chinese Academy of Sciences, Wuhan 430074, China
[d] College of Basic Medical Sciences, Hubei University of Chinese Medicine, 430000 Wuhan, China
[e] College of Landscape and Ecological Engineering, Hebei University of Engineering, Handan, China
[f] College of Forestry, Central South University of Forestry and Technology, Changsha 410004, Hunan, China

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ABSTRACT

The tung oil produced by the tung tree (*Vernicia fordii*) provides resources for the manufacture of biodiesel. Ubiquitin-specific proteases (UBPs) are the largest group of deubiquitinases and play key roles in regulating development and stress responses. Here, 21 *UBPs* were identified in *V. fordii*, roughly one-half the number found in *Manihot esculenta* and *Hevea brasiliensis*. Most *UBP* duplications are produced from whole-genome duplication (WGD), and significant differences in gene retention existed among Euphorbiaceae. The great majority of UBP-containing blocks in *V. fordii*, *V. montana*, *Ricinus communis*, and *Jatropha curcas* exhibited extensive conservation with the duplicated regions of *M. esculenta* and *H. brasiliensis*. These blocks formed 14 orthologous groups, indicating they shared WGD with *UBPs* in *M. esculenta* and *H. brasiliensis*, but most of these *UBPs* copies were lost. The *UBP* orthologs contained significant functional divergence which explained the susceptibility of *V. fordii* to *Fusarium* wilt and the resistance of *V. montana* to *Fusarium* wilt. The expression patterns and experiments suggested that *Vf03G1417* could affect the seed-related traits and positively regulate the seed oil accumulation. This study provided important insights into the evolution of *UBPs* in Euphorbiaceae and identified important candidate *VfUBPs* for marker-assisted breeding in *V. fordii*.

1. Introduction

Ubiquitination is involved in the regulation of almost all life activities such as DNA repair, signal transduction, cell-cycle progression, transcription regulation, gene expression, apoptosis, and endocytosis [1–4]. The ubiquitination of proteins is catalyzed by three sequential steps performed by E1, E2, and E3, respectively [5–7]. However, deubiquitination can remove ubiquitin from the target protein, thereby reversing ubiquitination [8,9]. Deubiquitinating enzymes (DUBs) can control the reversibility of the ubiquitination process and allow the ubiquitination mechanism to make changes, edit, or nixing decisions [10,11].

In plants, the deubiquitination enzymes contained five different types, of which ubiquitin-specific protease (UBP) is the largest subfamily [12–15]. Members of the UBP family can be determined by the ubiquitin carboxyl-terminal hydrolase (UCH) domain, which contains two highly conserved motifs [15,16]. There are the “cysteine (Cys)” and “histidine (His) boxes”, which are key parts of the catalytic sites and are proposed as part of the active sites of these thiol proteases [6,15,16]. UBP proteins not only contain the conserved UCH domain, but also other domains, such as Ubiquitin-associated (UBA), the domain present in Ubiquitin-Specific Proteases (DUSPs), the meprin and TRAF homology (MATH), the zinc finger domain ZnF-UBPs, and the MYND-type zinc finger domain (ZnF-MYND) domains [9,15]. To date, 27 *UBP* family members have been identified in *Arabidopsis thaliana*, and these genes have been further divided into fourteen sub-clades [16]. The *UBPs* perform multiple functions in plants, including pollen development and transmission, transcriptional regulation, signal transduction, mitochondria

* Correspondence to: Y. Cao, CAS Key Laboratory of Plant Germplasm Enhancement and Specialty Agriculture, Wuhan Botanical Garden, Chinese Academy of Sciences, Wuhan 430074, China.
** Corresponding authors.
E-mail addresses: xfcypeng@126.com, caoyunpeng@wbgcas.cn (Y. Cao), lzhangss@msn.com (L. Zhang), jianglanhi@163.com (L. Jiang).

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morphogenesis, root hair elongation, endoreplication, cell proliferation, and canavanine resistance [5,12,14,17–20].

Tung tree (*Vernicia* spp.) is one of the leading cultivated woody oil plants which is widely grown in subtropical areas, and its seeds are mainly used to produce tung oil, biomass diesel, or as raw materials for other chemical products [21]. The *Fusarium* wilt caused by *Fusarium oxysporum* f. sp. *fordiis* (*Fof*) caused devastating damage to the *Vernicia fordii* [22–24]. However, although the oil quality and yield of *V. montana* are relatively low, they have significant resistance to *Fusarium* wilt [22–25]. Due to its important economic value, integrative omics data on resistance to *Fusarium* wilt disease were recently sequenced and released [21,25], which are helpful for comprehensive analysis of *UBP* genes in tung tree. The main goal of this study is to compare all the *UBP* genes in tung tree with the other five Euphorbiaceae genomes and *A. thaliana* to calculate divergence times of gene duplication events, determine the phylogenetic relationships, and provide novel evidence for molecular evolution. By comparing the expression patterns of *UBPs* between *V. fordii* and *V. montana* after *Fusarium* wilt infection, we further speculated the functions of *UBPs* related to defense and resistance against *Fusarium* wilt disease.

2. Results and discussion

2.1. Phylogenetic tree divides UBPs into fourteen major clades

In our study, we identified 170 *UBPs* in six Euphorbiaceae genomes, including 21 in *V. fordii*, 26 in *V. montana*, 24 in *J. curcas*, 21 in *R. communis*, 34 in *M. esculenta*, and 44 in *H. brasiliensis*, with manual curation and strict filtering (Table S1). The numbers of *UBPs* in *M. esculenta* and *H. brasiliensis* were more than in *R. communis*, *J. curcas*, *V. montana*, and *V. fordii*, indicating that the members of the *UBP* family in *M. esculenta* and *H. brasiliensis* have a greater expansion. This might be because *M. esculenta* and *H. brasiliensis* shared the recent common WGD event, while the other four Euphorbiaceae species did not experience this event [26–32].

To further explore the phylogenetic relationships of *UBPs* in Euphorbiaceae genomes, all identified *UBPs* were aligned with 27 *AtUBPs*. Liu et al. (2008) have identified *AtUBPs* in *A. thaliana* and found that they play important roles in plant growth, development, and stress resistance [15]. We constructed the maximum likelihood (ML) tree and then revealed that all *UBPs* from Euphorbiaceae and *A. thaliana* could be divided into fourteen clades (C1-C14), using *HsUBP* (GenBank ID: NP_006528.2) from *Homo sapiens* as an outgroup, which was in agreement with those published papers by Yan et al. (2000) and Liu et al. (2008) [16,21]. At the tip of the clade (Fig. 1), the shorter branch length proved the strong amino acid conservation of its cluster genes and illustrated the close evolutionary relationship between the members. In each clade, a branch with more than one *UBP* gene from the same species was likely to have a gene duplication event, and the lack of representatives of some species might be due to a gene loss event. Each of the six Euphorbiaceae species and *A. thaliana* contributed at least one *UBP* in most clades, except for clade 6 and clade 12 (Table 1). Clade 6 contained *UBPs* from *A. thaliana*, *M. esculenta. H. brasiliensis*, *J. curcas*, *V. fordii*, and *V. montana*, and clade 12 contained *UBPs* from *A. thaliana*, *M. esculenta. H. brasiliensis*, *J. curcas*, *R. communis*, and *V. montana* (Table 1). The number of *UBPs* from *M. esculenta* and *H. brasiliensis* in most clades were more abundant than that of other Euphorbiaceae species. Members of different species in each clade might have evolved from a common ancestor gene through lineage differentiation and species formation. Therefore, the 14 clades defined here provided a basic framework for the subsequent determination of *UBP* paralogous genes and orthologous genes.

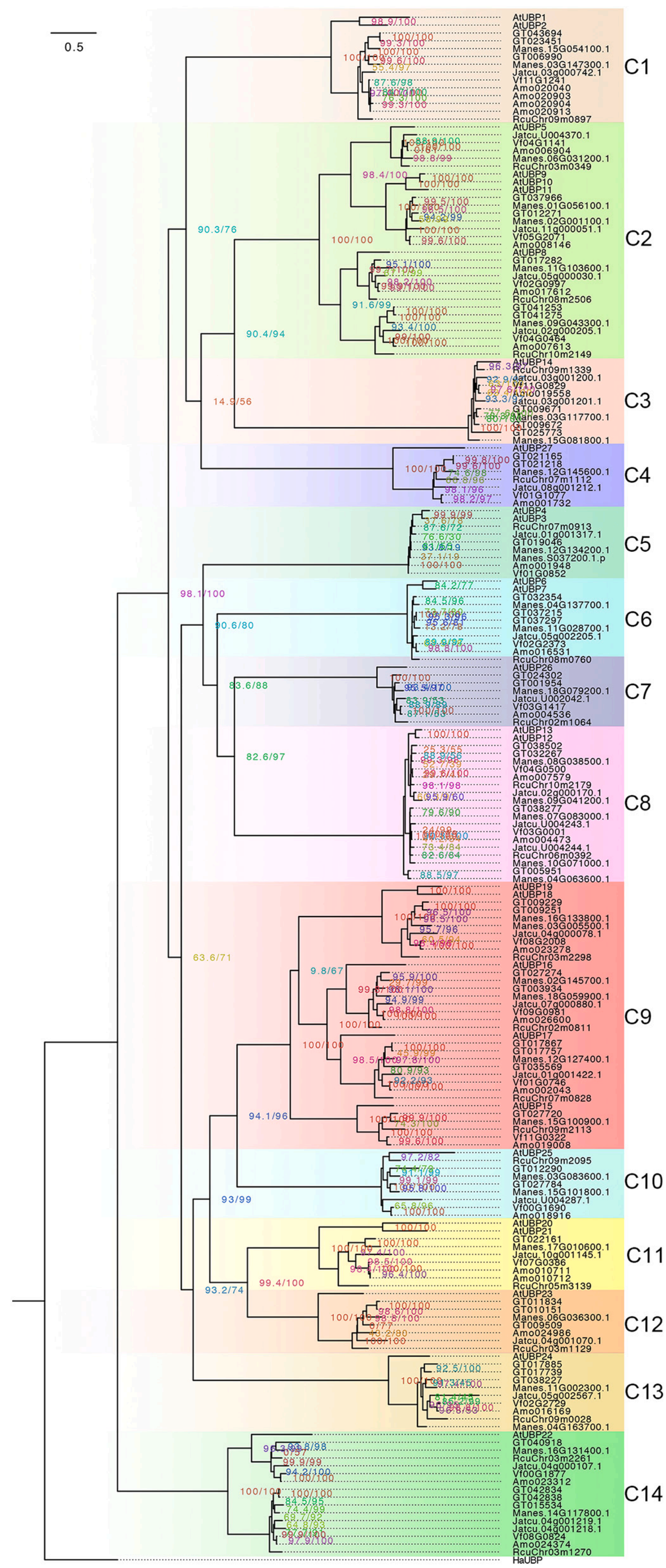

Fig. 1. Maximum likelihood tree of *UBP* genes in six Euphorbiaceae and *A. thaliana*. According to the *A. thaliana* homologs nomenclature, all *UBPs* from six Euphorbiaceae and *A. thaliana* were clustered into 14 clades (C1-C14) and were represented by different colors.

2.2. Expansion manners of UBPs within six Euphorbiaceae species

In order to examine the relationship between the genetic differences

Table 1
List of 14 UBP clades names and numbers of *A. thaliana* and Euphorbiaceae.

UBP clades	*A. thaliana*	*M. esculenta*	*H. brasiliensis*	*J. curcas*	*R. communis*	*V. fordii*	*V. montana*	Total
C1	2	2	3	1	1	1	4	14
C2	5	5	5	4	3	4	4	30
C3	1	2	3	2	1	1	1	11
C4	1	1	2	1	1	1	1	8
C5	2	2	1	1	1	1	1	9
C6	2	2	3	1	0	1	1	10
C7	1	1	2	1	2	1	1	9
C8	2	5	4	3	2	2	2	20
C9	5	6	8	3	4	4	4	34
C10	1	2	2	1	1	1	1	9
C11	2	1	1	1	1	1	2	9
C12	1	1	3	1	1	0	1	8
C13	1	2	3	1	1	1	1	10
C14	1	2	4	3	2	2	2	16

in the *UBP* gene family of each Euphorbiaceae plant and the corresponding expansion pattern, we further studied the gene duplication events of the *UBP* family among six Euphorbiaceae species. Euphorbiaceae species have experienced one or more WGD events, so the large-scale gene duplication events might play an important role in the evolution of the *UBP* gene family [26–32]. To investigate this possibility, we searched for gene similarity of *UBPs* flanking sequences. Subsequently, we defined the region as a syntenic block where four or more conserved homologs were located within 15 genes downstream and upstream between genomes [33].

In *R. communis*, eight conserved genes were found in the flanking sequence of *RcuChr02m0811/RcuChr07m0828*, suggesting that these genes were thought to have evolved from the large-scale duplication (Fig. 2 and Table S2). Only one gene pair (*Vf01G0746/Vf09G0981*) from *V. fordii* and one gene pair from *J. curcas* (*Jatcu.07g000880/Jatcu.01g001422*) were found to be involved in the large-scale duplication event, respectively. In *H. brasiliensis*, four pairs of genes, *GT009229/GT009251*, *GT037215/GT037297*, *GT011834/GT010151*, and *GT021165/GT021218*, were in the adjacent positions of the chromosomes, indicating that they should be produced by tandem duplications. The flanking genes of 15 gene pairs contained at least three conserved homologous genes, indicating that they should be the products of the large-scale duplications, such as *GT024302/GT001954*, *GT017885/GT038227*, and *GT017885/GT017739*. In *M. esculenta*, we did not detect tandem duplication of *UBPs*, but eight gene pairs were found in the duplication of chromosome segments (Fig. 2 and Table S2). In *V. montana*, we have not found any *UBPs* that originated from the large-scale duplication, only one gene pair (*Amo010711/Amo010712*) might be produced by tandem duplication.

The rate of synonymous substitution (Ks) at each site is constant over time, calculating the Ks of conserved flanking genes can be used to estimate the time of large-scale duplication events [34,35]. According to the Ks values of syntenic blocks flanking the *UBP* pair, these blocks of six Euphorbiaceae were divided into two groups (Table S2). The syntenic

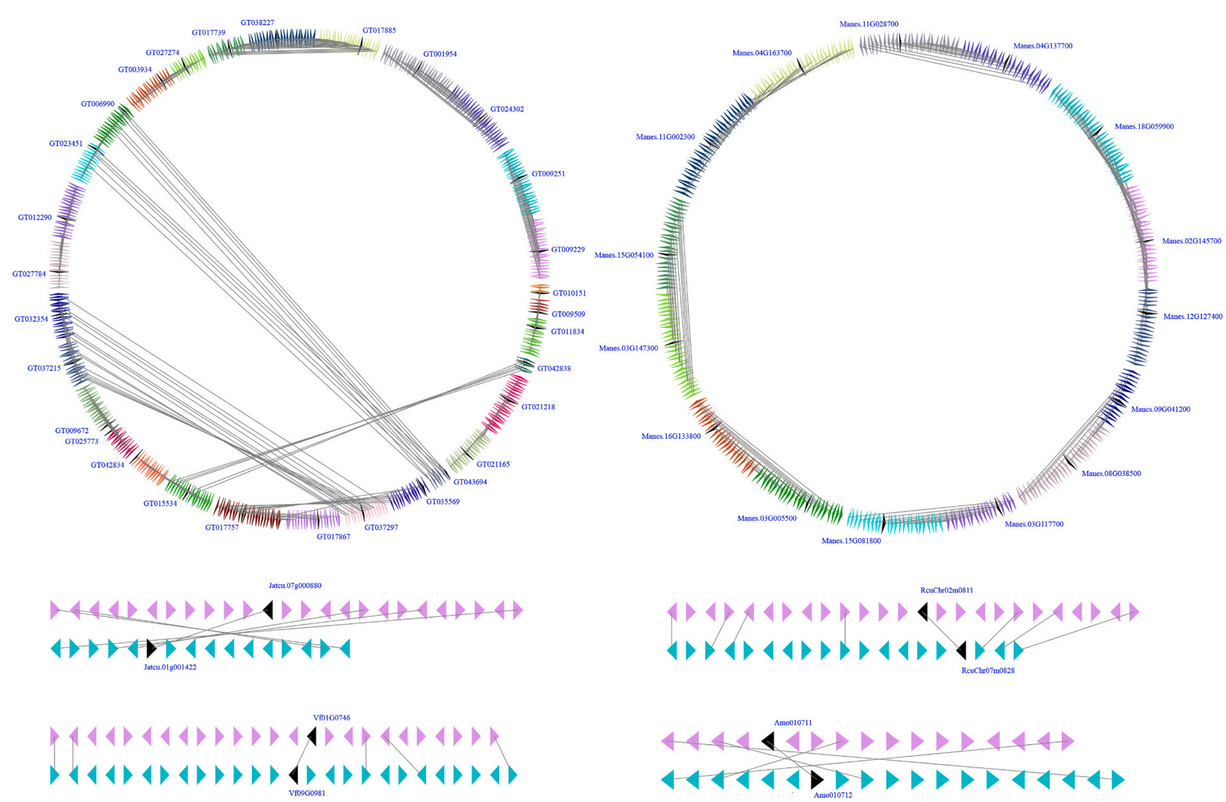

Fig. 2. Microsynteny related to *UBPs* in six Euphorbiaceae. Triangles represented a *UBP* gene and its adjacent flanking genes. The gene's orientation was represented by the triangle. The homologous genes were linked by gray lines between two fragments.

blocks flanking 19 *UBP* pairs contained the mean Ks value of 0.35, corresponding to a recent WGD event (Fig. 3a; ~15.3 Mya ago) [21]. In the other group, the syntenic blocks flanking four *UBP* pairs yielded the mean Ks values of 1.5, corresponding to an ancient WGD event (Fig. 3a; ~34.55 Mya ago) [21,32]. For *GT017757* and *GT035569*, the mean Ks value was 0.6, indicating that this pair originated between two rounds of WGD events (Table S2). However, the relationship between *GT043694* and *GT023451* was uncertain, speculating that they might originate from the recent segmental duplication event. These results indicated that two WGD events contributed to the expansion of *UBP* gene family members in the Euphorbiaceae species.

In the duplicated network of both *M. esculenta* and *H. brasiliensis UBPs*, when two duplicated genes produced by the recent WGD cannot be found simultaneously, we inferred that an ancient gene loss event may have occurred [36]. In the ancestor of both *M. esculenta* and *H. brasiliensis* lineages, an ancient WGD should have produced at least 24 *MeUBPs* and 14 *HbUBPs* (Fig. 4). However, twelve and six of these in *M. esculenta* and *H. brasiliensis* lineages lacked both copies, respectively, which would have been produced by the recent WGD, indicating that about 47 % of the ancient duplicates were lost during the long period of evolution. Additionally, the number of both *M. esculenta* and *H. brasiliensis UBPs* derived from the recent WGD should be 58 considering the ancient gene losses event. Among these, only one gene pair lost one copy of the gene, which suggests that only approximately 1.7 % of recent duplicates have experienced gene loss events during evolution. On the contrary, in *V. fordii*, *J. curcas*, and *R. communis*, only several UBP-containing segments were derived from WGDs. Where do the other *UBP* genes in these Euphorbiaceae species originate?

2.3. Massive losses of duplicated UBPs in R. communis, J. curcas, V. montana, and V. fordi

To trace the orthologous relationships and evolutionary origins of *UBPs* in Euphorbiaceae genomes, we used the *UBP* family members as anchor genes to determine the molecular evolutionary history of the chromosome regions where they are located. The relationships between syntenic orthologous genes of *UBPs* in six Euphorbiaceae genomes were exhibited in Fig. 3b-c, suggesting that strongly conserved microsynteny among these regions across *R. communis*, *J. curcas*, *V. montana*, *V. fordi*, *M. esculenta*, and *H. brasiliensis* was observed significantly. Subsequently, we could assemble 141 out of 170 UBP-containing flanking sequences from these six genomes into 14 groups (Fig. 3b-c and Table S3). A total of 141*UBPs* (19 from *V. montana*, 18 from *R. communi*, 39 from *H. brasiliensis*, 19 from *J. curcas*, 31 from *M. esculenta*, and 15 from *V. fordi*) were showed in the 14 orthologous segments groups. Fourteen ancient gene lineages deduced by phylogenetic analysis have been verified by *R. communis-J. curcas-V. montana-V. fordi-M. esculenta-H. brasiliensis* microsynteny, indicating that there is a one-to-one correspondence between ancient gene lineages and syntenic orthologous groups (Fig. 3b).

Segments of different Euphorbiaceae genomes within an orthologous group are thought to have shared the ancient WGD [37]. Only one region of *R. communis*, *J. curcas*, *V. montana*, and *V. fordi* in many groups was comparable to syntenic blocks of *M. esculenta* and *H. brasiliensis*, indicating that one member of the *UBP* pair produced from WGD was lost in their ancestral lineages. Additionally, the *UBPs* orthologous genes were usually scanned in the syntenic regions of *M. esculenta* and *H. brasiliensis*, and there were no counterparts in one genome of *R. communis*, *J. curcas*, *V. montana*, and *V. fordi*, suggesting that *UBPs* produced from WGD were lost in its ancestral lineage (Fig. 3c). For example, the *Jatcu.05g002205.1/Vf02G2373/Amo016531* anchored regions exhibited microsynteny with two duplicated regions of *M. esculenta* and *H. brasiliensis* containing *Manes.04G137700/Manes.11G028700* and *GT032308/GT037175* (Table S3). Significantly, we also found that the regions containing the orthologous *UBP* genes only contain microsynteny in two Euphorbiaceae genomes, indicating that their orthologous genes are lost in the other four Euphorbiaceae species. For example, the orthologous genes of *GT012290/GT027784/Manes.03G083600/Manes.15G101800* were missing in *R. communis*, *J. curcas*, *V. montana*, and *V. fordi*.

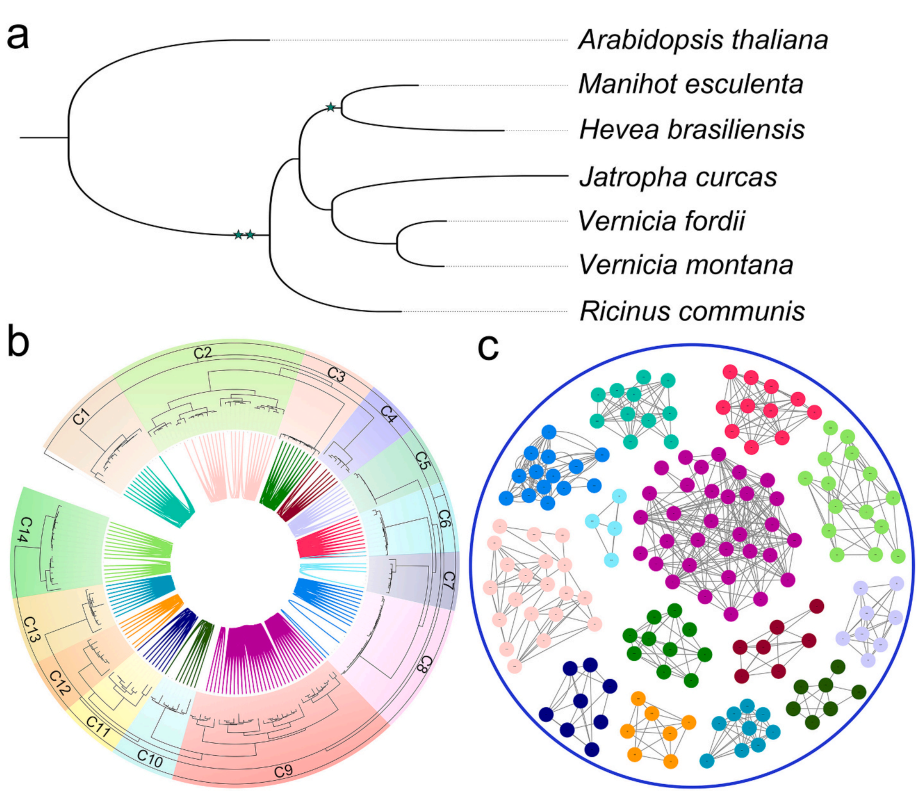

Fig. 3. The species tree and orthologous relationships for *UBPs* in six Euphorbiaceae. (a) The species tree of six Euphorbiaceae and *A. thaliana*. Orthofinder was used to identify single-copy orthologous genes, and IQ-tree was used to construct species tree. Green stars suggested WGD. (b) The orthologous network of the *UBPs* gene family using all the detected syntenic relations, and there is a one-to-one correspondence between ancient gene lineages and syntenic orthologous groups. (c) Synteny network for clades of *UBP* gene family. Communities were labeled by the clades involved from ML tree. The number of edges it has (node degree) corresponds to the size of each node.

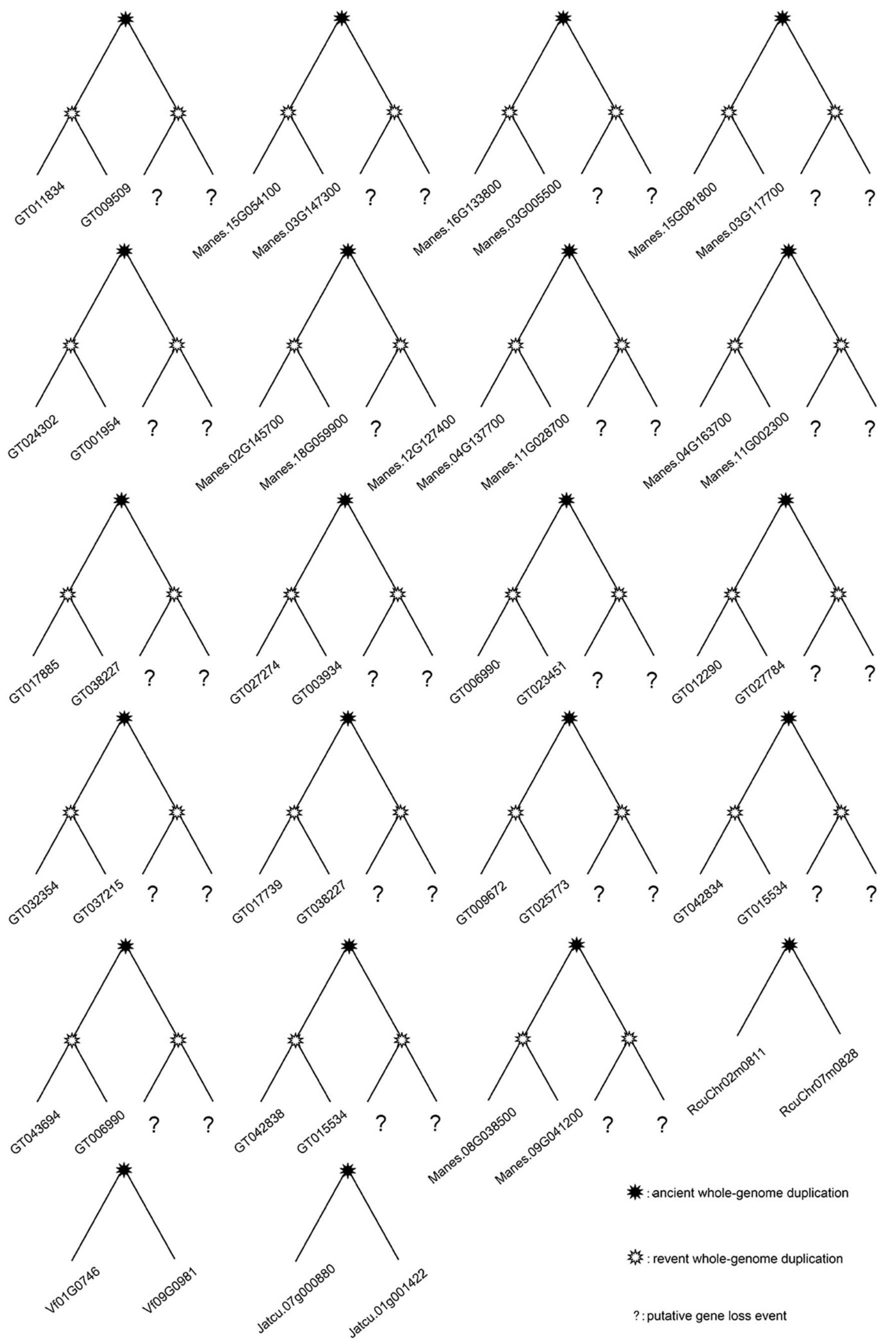

Fig. 4. Idealized gene trees of the duplicated *UBPs* groups in *V. fordii*, *V. montana*, *J. curcas*, *R. communis*, *M. esculenta*, and *H. brasiliensis.* Each tree indicated a duplication group from large-scale gene duplication. Every *UBP* of *M. esculenta* and *H. brasiliensis* was expected to be exhibit in four copies after two WGDs. Similarly, the number of UBPs in *V. fordii*, *V. montana*, *J. curcas*, and *R. communis* will have doubled after an ancient WGD.

2.4. Expression patterns of UBPs during Fusarium wilt infection between V. montana and V. fordii

The oil produced by *V. fordii* seeds is widely used in industry, while its growth and yield are severely affected by *Fusarium* wilt. In contrast, despite the relatively lower oil yield and quality of *V. montana*, it has significant resistance to *Fusarium* wilt [22–24]. Given that the *UBP* family is involved in the defense against pathogen infection [7,38,39], and that *V. fordii* is susceptible while *V. montana* is resistant, it is possible that members of *UBP* family might have experienced functional divergence with regard to disease tolerance during the evolution of *V. montana* and *V. fordii* to obtain specific functions. In our study, we compared the expression trend of *UBPs* between *V. montana* and *V. fordii* to determine the candidate *UBPs* involved in *Fusarium* wilt resistance. As shown in Fig. 5, both down-regulated and up-regulated expression patterns were reflected in *UBPs* in these two *Vernicia* species. Compared with *V. fordii* tree infected with *Fusarium* wilt, the *UBPs* in *V. montana* were significantly up-regulated during *Fusarium* wilt infection (Fig. 5), indicating that *UBPs* might play important roles in resistance to *Fusarium* wilt.

Orthologous genes that originated from a common ancestor may have similar functions. Using the Reciprocal best hit (RBH) method [40], we identified one-to-one orthologous gene pairs of *UBPs* between *V. montana* and *V. fordii*. As shown in Fig. 5, we detected 21 orthologs and found that most of these *UBPs* have diverse expression profiles in response to pathogen attacks between *V. montana* and *V. fordii*.

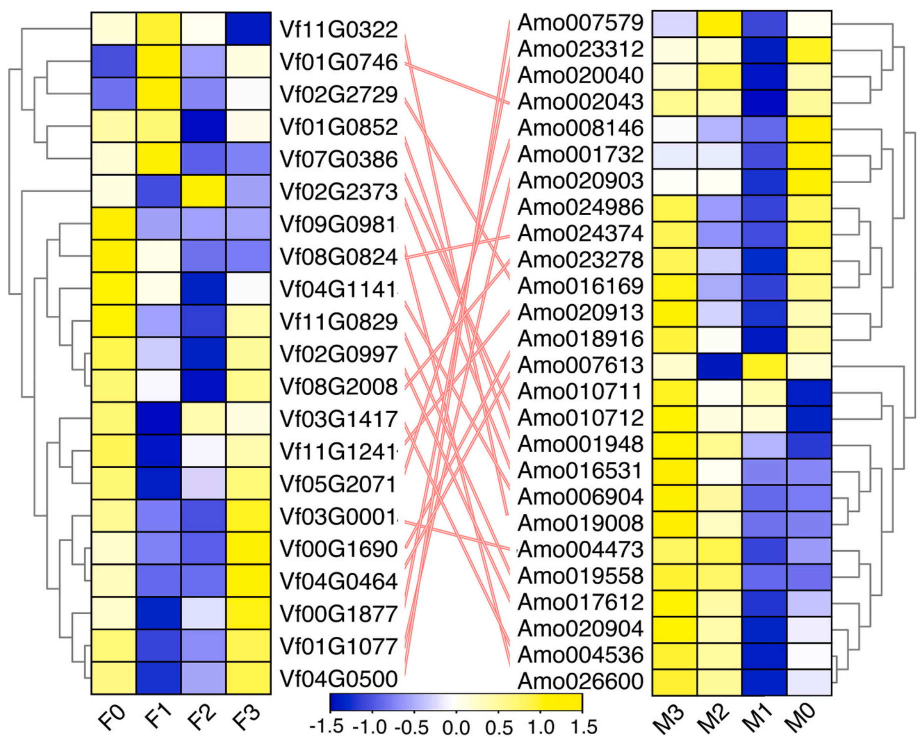

Fig. 5. Expression profiles of *UBPs* under *Fusarium* wilt infection in two *Vernicia* species. M0-M3 suggested the expression of *UBPs* in *V. montana* during the infection stage (0, 1, 2, 3) by the pathogen *Fusarium* wilt, and F0-F3 suggested the expression of *UBPs* in *V. fordii* during the infection stage (0, 1, 2, 3) by the pathogen *Fusarium* wilt. The one-to-one orthologous relationships of UBPs were represented by the yellow lines between *V. montana* and *V. fordii*.

Subsequently, the expression correlations were further calculated to detect the degree of expression diversity between orthologous gene pairs in these two *Vernicia* species. Finally, only three *UBP* orthologous gene pairs are ongoing divergent, including *Vf08G2008/Amo023278*, *Vf05G2071/Amo008146*, and *Vf00G1877/Amo023312*. A total of 18 pairs were found to be divergent, while no *UBP* orthologous gene pairs were non-divergent (Table 2). Additionally, the analysis of the branch-site model suggested no significant positive selection on codon sites of any orthologous gene pairs between *V. montana* and *V. fordii*, indicating that 85.7 % (eighteen out of twenty-one) orthologous gene pairs occurred functional divergence. According to the Ks analysis, we found that most *UBP* orthologous gene pairs were very old and divergent (Table 2), and were later than the species differentiation time of *V. montana* and *V. fordii*. Summarizing the above analysis, we found that the *UBP* orthologous genes evolved from the same ancestor have significant functional divergence, which might explain the susceptibility of *V. fordii* to *Fusarium* wilt and the resistance of *V. montana* to *Fusarium* wilt.

2.5. Expression patterns of UBPs at different developmental stages of V. fordi

Members of the *UBP* family play diverse roles during developmental stages and in different tissues, which can be clearly exhibited from their

Table 2
Divergence between orthologous *UBP* gene pairs between *V. montana* and *V. fordii.*

Gene1	Gene2	Ka	Ks	Ka/Ks	r	Gene expression
Vf02G2729	*Amo016169*	0.0184	0.0501	0.3674	−0.4895	Divergent
Vf01G0746	*Amo002043*	0.0157	0.0227	0.6939	−0.4056	Divergent
Vf07G0386	*Amo010712*	0.0118	0.0471	0.2495	−0.3846	Divergent
Vf04G1141	*Amo006904*	0.0093	0.0352	0.2639	−0.3706	Divergent
Vf11G0322	*Amo019008*	0.0247	0.0569	0.4347	−0.3566	Divergent
Vf00G1690	*Amo018916*	0.0123	0.0289	0.4244	−0.1608	Divergent
Vf09G0981	*Amo026600*	0.0156	0.0326	0.4798	−0.1329	Divergent
Vf01G1077	*Amo001732*	0.01	0.0212	0.4729	−0.1259	Divergent
Vf11G0829	*Amo019558*	0.0108	0.0263	0.4101	−0.1119	Divergent
Vf02G0997	*Amo017612*	0.0136	0.0166	0.8186	−0.021	Divergent
Vf04G0464	*Amo007613*	0.0205	0.0274	0.7484	−0.014	Divergent
Vf01G0852	*Amo001948*	0.0012	0.053	0.0218	0.08392	Divergent
Vf03G0001	*Amo004473*	0.0012	0.0292	0.0396	0.09091	Divergent
Vf03G1417	*Amo004536*	0.0118	0.0299	0.3952	0.18881	Divergent
Vf02G2373	*Amo016531*	0.0094	0.021	0.4462	0.2028	Divergent
Vf08G0824	*Amo024374*	0.0015	0.0566	0.026	0.2028	Divergent
Vf04G0500	*Amo007579*	0.0032	0.0451	0.0702	0.25874	Divergent
Vf11G1241	*Amo020913*	0.0081	0.0282	0.2853	0.26573	Divergent
Vf08G2008	*Amo023278*	0.0179	0.0367	0.4882	0.37762	Ongoing divergent
Vf05G2071	*Amo008146*	0.0051	0.0266	0.192	0.38462	Ongoing divergent
Vf00G1877	*Amo023312*	0.0238	0.0447	0.5319	0.41958	Ongoing divergent

expression profiles in other plants [12,16,41]. The expression patterns from the RNA-Seq dataset could generate different expression profiles for each *UBP*, providing important candidate genes for future *UBP* functional verification and applications. *UBPs* have higher expression abundance in flowers, seeds, and pollen in *A. thaliana* and *O. sativa*, suggesting that they play important roles in the development of reproductive tissues [13,15]. In our study, the *UBPs* exhibited tissue-specific expression and differential expression patterns (Fig. 6). The *UBPs* from Class A showed high expression in M1 (male flowers at 30 days before flowering), the *UBPs* from Class B exhibited high expression in both male and female flower development, while the *UBPs* from Class C showed high expression during seed development. Indeed, *AtUBP12* and *AtUBP13* are highly expressed in *A. thaliana* flowers and have been shown to play a role in the regulation of the circadian clock and photoperiodic flowering. Remarkably, *Vf03G0001* and *Vf03G1417* were highly expressed during seed development, and *Vf03G1417* was clustered into the same clade with *AtUBP26*. It has been confirmed that *AtUBP26* is required for *A. thaliana* seed development [42]. *Vf11G0322* and *Vf09G0981* were highly expressed in the S1 (seeds at 10 weeks after flowering), and *Vf11G0322* was clustered into the same clade with *AtUBP15*. It has been confirmed that *OsUBP15* and *AtUBP15* play an important role in regulating seed and organ size [43,44]. Therefore, we deduced that *Vf03G1417* and *Vf11G0322* might play key roles in controlling seed and organ size.

2.6. Heterogeneous expression of Vf3G1417 in A. thaliana

The functions of *UBPs* have been widely studied and reported in some model plants. For example, *AtUBP14* is required for early embryo development in *A. thaliana* [45]. *FLA*, which is a homolog of *UBP*, is essential for the development of palea and lemma in *Oryza sativa* [46]. Based on the expression patterns of *VfUBPs*, *Vf3G1417* was expressed predominantly in developing seeds (Fig. 7a), indicating this gene might regulate seed-related traits. Therefore, we further studied the function and role of *Vf3G1417* in controlling seed development. In general, *A. thaliana* seed oil contains several major fatty acids, rachidic acid (C20:0), cis-11-eicosenic acid (C20:1Δ9), eleostearic acid (C18:3Δ9, 11, 13), linolenic acid (C18:3Δ9, 12, 15), linoleic acid (C18:2Δ9, 12), oleic acid (C18:1Δ9), stearic acid (C18:0), cis-9-hexadecenoate (C16:1Δ9), and palmitic acid (C16:0). To test whether *Vf3G1417* can affect the seed-related traits, the oil content was measured in seeds of wild-type and over-expression plants (Fig. 7b). We only found no obvious differences in seed coat colour between mature seeds of wild-type and over-expression plants. However, these seed morphological traits, including dry weight, seed size, and oil contents, increased significantly in over-expressed seeds than in wild-type seeds (Fig. 7b). These data showed that *Vf03G1417* not only played an important role in the seeds growth and development but also positively regulated the seed oil accumulation.

3. Conclusion

In our study, we provide an overview of *UBPs* in six Euphorbiaceae genomes, including their phylogenetic relationships, microsynteny, and expression patterns. According to these findings, the evolutionary history of ancestral *UBPs* was tracked among these six Euphorbiaceae genomes. Both phylogenetic relationships and microsynteny suggested that massive losses of duplicated *UBPs* have occurred in *R. communis*, *J. curcas*, *V. montana*, and *V. fordi*. Based on the comparison of *UBPs* between *V. montana*, and *V. fordi*, the transcriptional expression, and function of several genes have become clearer, indicating that the *UBP* orthologous genes evolved from the same ancestor containing significant functional divergence which further explains the susceptibility of *V. fordii* to *Fusarium* wilt and the resistance of *V. montana* to *Fusarium*

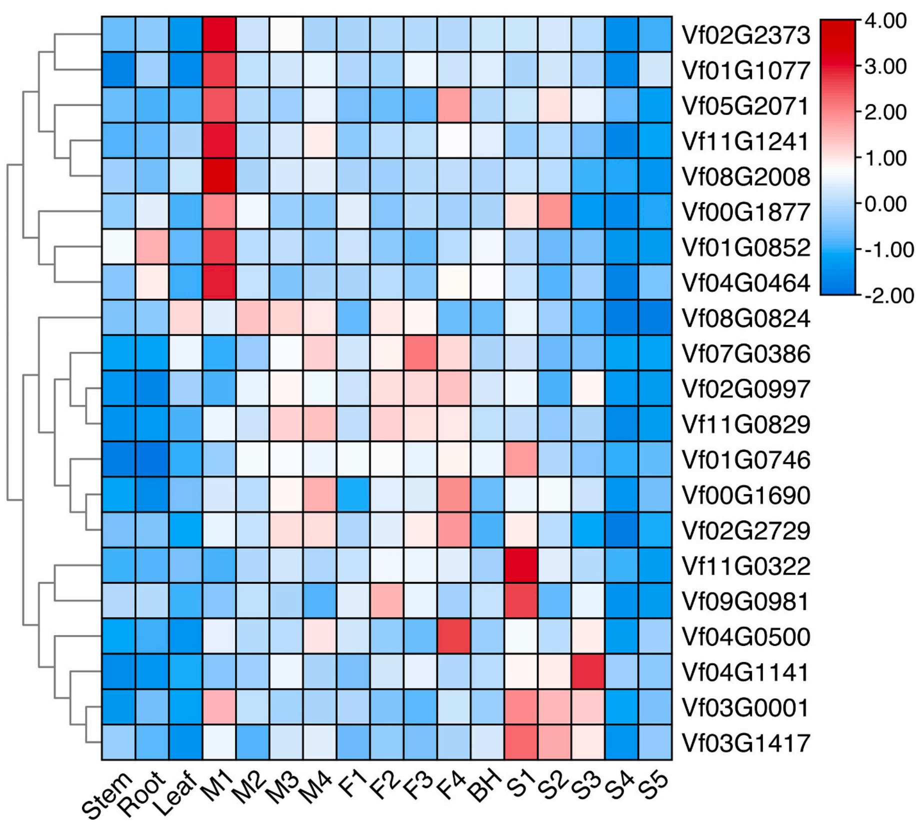

Fig. 6. Expression profiles of *UBPs* in 17 different *V. fordii* tissues. Numbers on the x-axis indicated the following: hermaphrodite (BH), female flowers at 30, 20, 10, and 1 days before flowering (F1-F4), male flowers at 30, 20, 10, and 1 days before flowering (M1-M4), and seeds at 10, 15, 20, 25, and 30 weeks after flowering (S1-S4).

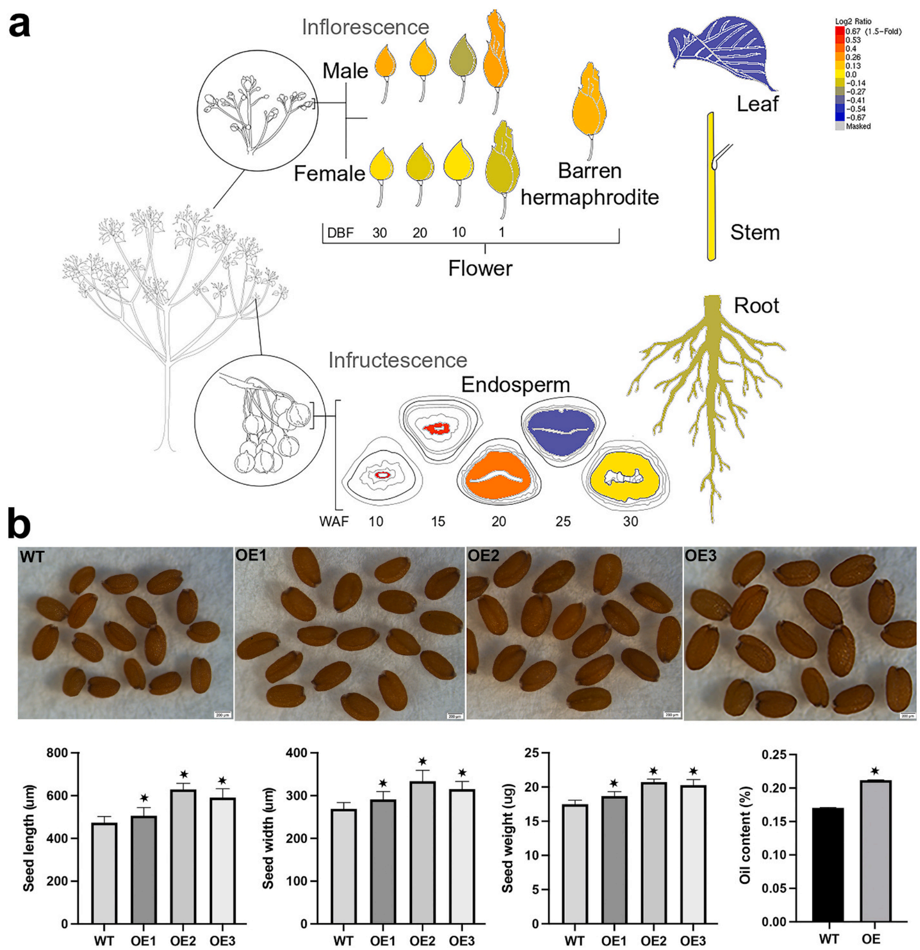

Fig. 7. Comprehensive characterization of over-expression *Vf03G1417* seeds. (a) Expression profile of *Vf03G1417* in 17 different *V. fordii* tissues. DBF indicated days before flowering, and WAF suggested weeks after flowering. (b) Microscopic observation and quantitative comparisons of mature seeds randomly selected from WT and OE plants. Quantitative comparisons of oil content, seeds length, seeds width and seeds dry weight between WT and OE plants.

wilt. The expression patterns of *UBPs* at different tissues or developmental stages of *V. fordi* indicating that the functions of *UBPs* were associated with seed development. Further experiments confirmed that *Vf03G1417* could affect the seed-related traits and positively regulate the seed oil accumulation. Our study has not only traced the evolution of *UBPs* in Euphorbiaceae genomes, but also for determining the genetic mechanisms of resistance to *Fusarium* wilt infection, which will provide valuable clues for marker-assisted breeding in *V. fordii*.

4. Methods

4.1. Data sources and sequence retrieval

In our study, multiple steps were carried out to determine the *UBP* genes from Euphorbiaceae species. The castor bean (*Ricinus communis*), *V. fordii*, *V. montana*, rubber tree (*Hevea brasiliensis*) sequences with GFF files were downloaded from the NCBI database [26–32]. The physic nut (*Jatropha curcas*) and cassava (*Manihot esculenta*) sequences with GFF files were obtained from Jatropha genome database and Ensembl database, respectively [26–32]. The *A. thaliana UBPs* were obtained from the TAIR database based on previously reported articles [15,16]. Subsequently, the *A. thaliana UBPs* were used as queries for a BlastP search against the local Euphorbiaceae genomes. Finally, the potential members of the UBP family were further verified by PFAM [47], InterPro [48], and SMART [49] databases. If there are obvious errors in the sequences or missing UCH domains, they will be discarded manually.

4.2. Phylogenetic trees construction

The amino acid sequences of UBP proteins in six Euphorbiaceae and *A. thaliana* were aligned with the aid of MAFFT [50]. We constructed the phylogenetic tree by IQ-tree with maximum likelihood (ML) method, based on the best model as implemented in IQ-tree [51]. Both bootstrap tests with 10,000 replicates and an SH-aLRT analysis with 1000 random addition replicates were carried out to calculate the reliability of the ML tree. iTOL (https://itol.embl.de/) and Figtree (http://tree.bio.ed.ac.uk/) were performed to visualize the phylogenetic trees [52].

4.3. Microsynteny analysis

To identify the expansion of the *UBPs*, we obtained the physical locations of these genes from GFF files. If multiple gene family members were located in the same or surrounding interval regions in the genome, they were considered to have been tandem duplications. For the identification of large-scale duplication events, we used a method similar to Cao et al. (2019) and Maher et al. (2006) [34,53]. If the two *UBPs* were located in a syntenic block, their flanking sequences must be highly similar at the amino acid level [54]. We first located all *UBP* members on the genome and used them as anchor genes. Subsequently, 15 flanking sequences downstream and upstream of each anchor gene were analyzed by pairwise BlastP analysis, so as to identify whether there are duplication genes between the two independent regions. If at least three gene pairs are detected as having syntenic between the two regions, they were considered to have evolved from large-scale duplication events, as described by Cao et al. (2021) [33]. ParaAT2.0 was used to convert the gene and peptide files into AXT format [55]. The KaKs_Calculator 2.0 was used to carry out the Ka/Ks, Ks, and Ka values with the procedure of Nei and Gojobori (1986) [56,57].

4.4. RNA-seq analysis

The RNA-seq reads were downloaded NCBI (Accession numbers: PRJNA483508, PRJNA318350, and PRJNA445068). We used the pipeline Fastq clean to process and delete the low-quality base-calls of raw data [58]. The pipeline HISAT2 was used to map clean reads to the corresponding reference genome of *V. fordii* and *V. montana* with default parameters, respectively [59,60]. The StringTie was used to calculate the counts of the reads, and they were normalized using Fragments Per Kilobase per Million (FPKM) with default parameters [59], as described by Jiang et al. (2022) and Li et al. (2022) [61,62].

4.5. Expression correlation of orthologs UBPs between V. montana and V. fordii

We manually gathered the expression profiles of orthologs *UBPs* from our RNA-Seq data. Pearson's correlation coefficient (r) was implemented to investigate the similarity between expression patterns of each orthologous gene pair [5,63,64]. Significant values were performed to exhibit the degree of expression diversity: $r \geq 0.5$ indicates non-divergence, $0.3 \leq r < 0.5$ suggests ongoing divergent, and $r < 0.3$ means divergence [5,63,64].

4.6. Cloning and A. thaliana transformation of Vf03G1417

The coding sequence (CDS) of *Vf03G1417* was cloned from the *V. fordii* seeds using the primers: ATGTCTTCAAATCATGTTTTCC and TTAAACAACTTGTGATGAAATATC, and inserted into the PMDC32 overexpression vector by the Gateway cloning system. The pMDC32::Vf03G1417 was transformed into *A. thaliana* by agrobacterium-mediated the floral dip method. Transformants were selected in a media with hygromycin-containing medium (50 mg/ml) and then confirmed by PCR. Subsequently, we selected the homozygous T3 lines for further analysis.

4.7. Oil content analysis

We dried the T3-generation *A. thaliana* seeds to constant weight in an oven at 60 °C and then ground them into powder. Weigh 0.5 g of powder sample, added 10 ml of CH_3OH-NaOH (1.25 mol/L) solution to it, and mixed well. Subsequently, 1 ml of ice-cold acetic acid and 2 ml of n-heptane were added to the mixture for methyl esterification. We collected 1 ml of supernatant for fatty acid detection using gas chromatograph GC-2014 (Shimadzu, Japan), as described by Chen et al. (2010). We used the fatty acid methyl ester standards as references, including arachidic acid (C20:0), cis-11-eicosenic acid (C20:1Δ9), eleostearic acid (C18:3Δ9, 11, 13), linolenic acid (C18:3Δ9, 12, 15), linoleic acid (C18:2Δ9, 12), oleic acid (C18:1Δ9), stearic acid (C18:0), cis-9-hexadecenoate (C16:1Δ9), and palmitic acid (C16:0) (Sigma, St. Louis, MO).

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijbiomac.2022.08.163.

List of abbreviations

DUBs: Deubiquitinating enzymes, UBP: ubiquitin-specific protease, ML: maximum likelihood, Ks: synonymous substitution, WGD: whole-genome duplication.

Ethics approval and consent to participate

Permissions were not required to collect the samples in this study.

Consent for publication

Not applicable.

Availability of data and materials

Expression data of *Vernicia fordii* and *V. montana* were available in the NCBI SRA database with accession numbers: PRJNA318350, PRJNA483508, and PRJNA445068.

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CRediT authorship contribution statement

YPC conceived and supervised the research. YPC, YLL, LHW, and LJ designed the experiments. YPC and LZ performed the experiments and analyzed the results. YPC wrote the paper. All authors have read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no competing interests.

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